

Incentives or Symbols? Out-of-Distribution Cooperation in Multi-Agent LLM Systems

by

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Abstract

Large language model (LLM) agents have been shown to be sensitive to symbolic cues in prompts. This work investigates whether LLM agents in multi-agent environments reason over underlying incentive structures or rely on surface-levels symbolic cues. We introduce a structured benchmark based on a gridworld implementation of the Stag Hunt game, where agents must coordinate spatially and temporally to achieve high-reward outcomes. By comparing natural semantic descriptions (e.g., “stag” and “hare”) with neutral symbols (e.g. “ijjhu,” “guydguug”), we evaluate the robustness and generalization of zero-shot LLM agents across identical payoff structures. We design a multi-agent framework incorporating observations, memory, communication, and reputation, and evaluate five models across multiple experimental settings. Our evaluation decomposes agent behavior into multiple metrics, including reward efficiency, coordination dynamics, communication effectiveness, and reasoning quality. Results show that agent performance is highly sensitive to symbolic cues, with degradation under neutral framing. It suggests that there is a strong reliance on semantic priors rather than abstract reasoning. Furthermore, we examine reflection-based agents and find that while reflection improves behavioral consistency and long-term coordination, it does not consistently increase short-term rewards. These results demonstrate that current LLM agents lack robust generalization in multi-agent environments and remain highly sensitive to prompt semantics. This has important implications for the reliability of LLM-based agents in real-world decision-making tasks, where surface-level linguistic variations may lead to significant behavioral differences.

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1 Introduction

Large language models (LLMs) have recently been studied in multi-agent game settings to explore emergent social behaviors when multiple agents interact [Yao et al., 2023, Guo et al., 2024, Huang et al., 2024, Zhu et al., 2025]. Prior work shows that different forms of scaffolding, such as communication channels, memory, or role constraints, shape agent behaviors [Wang et al., 2023]. In controlled psychological and behavioral games, studies have explored whether agents cooperate, negotiate, or follow norms. For instance, Jin et al. [2024], Backmann et al. [2025] demonstrate that moral framing can shift strategies in social dilemmas, and agents exhibit human-like trade patterns in cooperative tasks [del Rio-Chanona et al., 2025]. Adding communication or sanctioning mechanisms has also been shown to increase prosocial efficiency in group settings [Sreedhar et al., 2025]. Agapiou et al. [2022] introduce a framework for testing social generalization across diverse multi-agent scenarios. However, in some settings, reasoning-oriented models still exhibit undesirable behaviors such as free-riding in public-goods games. Overall, LLM agents appear capable of achieving high performance in standard cooperation settings, particularly when supported by structured prompts and interaction mechanisms.

Despite this progress, a central question remains unresolved: **do LLM agents rely on task-specific strategies when familiar semantic cues are present, or do they possess meta-level reasoning abilities that enable them to infer incentives, communicate effectively, and coordinate in novel social environments?** This concern has been raised across recent studies showing that LLM behavior is highly sensitive to prompt phrasing, labels, and narrative framing, even when underlying payoff structures remain identical [Min et al., 2022, Zheng et al., 2023, Wang et al., 2023]. In game-theoretic settings, multiple studies find that LLM agents often base their decisions on familiar words, labels, or narratives in the prompt, rather than reasoning directly from the underlying payoff rules [Piatti et al., 2024, Backmann et al., 2025, Huang et al., 2025]. However, it remains unclear whether such behavior reflects lack of reasoning over incentives or reasoning conditioned on familiar symbolic contexts. Moreover, previous evaluations lack structured behavioral dimensions. Most studies rely on free-form dialogue analysis or aggregate rewards as primary metrics, overlooking latent behavioral properties or robustness [Park et al., 2023, Guo

et al., 2024, Zhu et al., 2025, Piedrahita et al., 2025, Huang et al., 2025].

This thesis addresses these gaps by introducing a benchmark based on the physical version of Stag Hunt game, which incorporates spatial and temporal dynamics. We use this extended version of the Stag Hunt game as a specific substrate and an example, designed to explicitly test generalization across models, representations, and environmental parameters. Agents must navigate a shared environment, coordinate actions over time, and manage partial observability. Rather than naming games or explicitly introducing entities (e.g., “stag” or “hare” in a staghunt game), environments are specified only through formal rules and payoff descriptions. Agents receive no task-specific training and must infer coordination incentives directly from textual rules and observations. We evaluate LLMs on games with similar underlying rules but different surface-level naming and framing. By varying the semantics while holding the structure constant, we test whether LLMs rely on contextual reasoning or simply respond to familiar language cues. In addition, we provide insights into how agents’ own reflections affect future behaviours. Moreover, unlike prior work that evaluates multi-agent LLMs using aggregate rewards or informal behavioral descriptions, our evaluation design explicitly decomposes social behavior into structured, quantitative metrics. These metrics capture coordination dynamics, robustness to semantic framing, and the alignment between agent actions, communication, and reasoning. It enables a principled test of generalization beyond familiar language cues. Through these approaches, the thesis aims to provide a clearer characterization of how and when LLM agents generalize social reasoning across symbolic contexts, game environments, and models.

2 Literature Review

Recent work has extended LLM agents beyond single-agent planning to multi-agent, game-theoretic, and socially interactive settings. Frameworks such as ReAct [Yao et al., 2023] and CAMEL [Guo et al., 2024] demonstrate that LLMs can coordinate, negotiate, and divide labor given structured prompting and communication channels. Building on this, subsequent studies have explored more complex scenarios, including economic games, social dilemmas, and collaborative simulations [Huang et al., 2024, Zhu et al., 2025]. They find that LLM agents can

reproduce qualitative patterns of human social behavior such as reciprocity, conditional cooperation, and norm enforcement. These capabilities are often scaffolded by mechanisms such as memory modules, persona constraints, or role-based conditioning, which bias strategies toward stable interaction dynamics [Park et al., 2023, Sreedhar et al., 2025]. Recent efforts have also examined moral reasoning and alignment in LLM agents. They show that models fine-tuned with value-based objectives can generalize moral strategies across dilemmas [Tennant et al., 2024], though ethical behavior often collapses when payoff incentives conflict with norms [Backmann et al., 2025]. In decentralized simulations, studies observe that LLM agents can be swayed by minorities [Ashery et al., 2025].

Though LLM agents have demonstrated strong capabilities in social and strategic interactions, zero-shot out-of-distribution (OOD) generalization remains a key challenge. Recent work has specifically examined how well LLM-based agents cooperate in multi-agent scenarios without re-training. For example, Smith et al. [2025] introduce the Concordia evaluation framework, which tests agents in a suite of mixed-motive games with unfamiliar partners and rules. Their findings reveal significant generalization gaps: agents that performed well on training-like negotiations often failed in more complex coordination tasks requiring persuasion or norm enforcement [Smith et al., 2025]. These findings are consistent with prior work showing that agent performance depends heavily on prompt design, memory mechanisms, and system-level scaffolding [Wang et al., 2023]. In language-driven settings, the problem is exacerbated by the open-ended nature of communication. LLM agents still exhibit brittle behavior when faced with novel game structures or partner strategies not encountered during development [Shinn et al., 2023, Li et al., 2023, Piatti et al., 2024]. Despite growing empirical evidence of these limitations, existing studies lack a systematic understanding of which specific social abilities generalize, such as when to communicate, what to communicate, and how communication supports coordination and in-context adaptation in novel environments.

Another line of research has explored sensitivity to symbolic and linguistic framing in LLM agent behavior. Minor changes in wording or context might lead the same model to produce different decisions, raising concerns about stability and alignment. Early work such as [Brown et al., 2020]

show that LLMs strongly depend on natural language cues and prompt structure for task performance. [Piatti et al., 2024] suggest that LLM agents can achieve sustained cooperation and form social norms in multi-agent environments, but such cooperation is sensitive to language, prompting, and environmental conditions. More recently, Zhang et al. [2025] demonstrate that across various tasks, simply reframing a prompt from a positive to a negative form (e.g. asking a yes/no question in the affirmative vs. the negative) caused large shifts in model responses, even though human judgments remain unchanged. In social decision-making, Moon et al. [2026] show that LLMs can reproduce known human biases by altering the textual instructions and context given to an LLM agent. They observed significant changes in generosity and trust, sometimes larger than the effects of demographic differences in real human experiments [Moon et al., 2026]. Similarly, Robinson and Burden [2025] vary the narrative framing of identical game-theoretic dilemmas and find LLM decisions to be sensitive to how the scenario is described, even when the underlying payoffs remain constant. These studies suggest that linguistic framing effects such as the choice of words to the narrative context can strongly influence LLM agents’ cooperation and moral decisions.

In addition, prior work has explored the use of self-reflection as a mechanism for improving LLM agent performance through iterative reasoning and memory accumulation. For example, [Shinn et al., 2023] introduce a framework where agents generate natural language reflections on past failures and incorporate them into subsequent decisions, and it shows performance gains without parameter updates. Similarly, [Park et al., 2023] demonstrates that reflection, combined with memory and planning, allows agents to form coherent long-term behaviors and interact in socially consistent ways. In embodied environments, [Wang et al., 2023] use reflection to refine strategies and support open-ended exploration. Despite these advances, those work primarily focuses on single-agent tasks or open-ended social simulations, leaving the role of reflection in structured multi-agent settings, particularly under incentive structures, unexplored.

In response to these issues, there are current benchmarks and methodologies to evaluate robustness to prompt variation, symbolic noise, and partner diversity in multi-agent settings. Frameworks like SOTOPIA [Zhou et al., 2023] enable rigorous testing of LLM agents in interactive games and

social scenarios with varied contexts. The Concordia contest itself serves as a benchmark for zero-shot cooperation robustness, evaluating agents when they face new partners and dialogue contexts [Smith et al., 2025]. Other work has introduced targeted stress-tests for semantic perturbations. For example, Pan et al. [2023] present the MACHIAVELLI benchmark, a suite of interactive text adventure games that force agents to make moral trade-offs, which evaluates whether models adhere to ethical norms under tempting rewards. There are also efforts to quantify an agent’s tolerance to prompt rephrasing and irrelevant symbolic noise. For instance, Liu et al. [2023] include in their AgentBench evaluation tests where functionally identical prompts are varied in wording, and they measure drops in task performance. Likewise, multi-agent negotiation studies have examined how different linguistic framings of a partner’s intent or the game’s goals can shift an LLM agent’s willingness to cooperate [Mukobi et al., 2023]. Across these benchmarks, the common aim is to stress-test LLM agents under variations in language and partner behavior, revealing whether agents rely on cues or truly understand the game.

This work differs from prior benchmarks by focusing on the Stag Hunt game to provide a testbed for symbolic sensitivity and generalization. We introduce a structured benchmark built around a social dilemma with systematic variations in linguistic cues and configurations. This design allows us to probe zero-shot generalization across models and settings to understand whether LLM agents truly understand underlying game dynamics or overfit to prompt cues. By targeting symbolic sensitivity in a minimal yet representative social setting, our approach complements broader evaluation suites and contributes a focused diagnostic tool for studying generalization failures in cooperative reasoning.

3 Agent and Environment Design

3.1 LLM Agents

Each agent is in essence a language model that receives system prompt, environment observations, and information from surrounding agents, and it outputs the action to take. They then follow their decisions and take a step. After all agents take actions, the environment then updates its states, which are then fed back into the agents as new observations and information for the next turn.

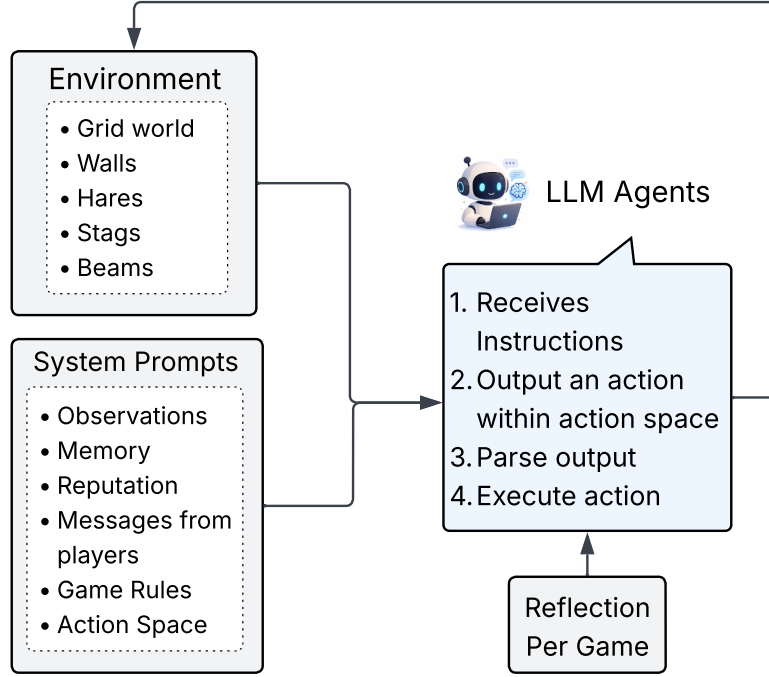


Figure 1: System Design Framework.

Moreover, each agent keeps an episodic memory about messages they have received and rewards so far to guide decision making. This section details agent configurations.

3.1.1 Action

All agents have six choices of actions as shown in [Table 1](#).

3.1.2 System Prompt

The general system prompt for agents are shown in [Figure 10](#).

3.1.3 Reward

`{reward_rule}` contains information about how agents can get reward through solo hunting or collaborating with other agents depending on the game environment. It might also provide some tips to guide decision-making. An example of reward rule for staghunt game is in [Figure 11](#).

Action ID	Action	Description
0	STAY	Agent does not move.
1	UP	Agent move to one block up its current position and face up.
2	RIGHT	Agent move to one block to the right of its current position and face right.
3	DOWN	Agent move to one block down its current position and face down.
4	LEFT	Agent move to one block to the left of its current position and face left.
5	ATTACK	Agent does not move and release a beam to attack potential resource in front of it.

Table 1: Action Descriptions.

3.1.4 Observation

In each turn, i.e. when all agents make a decision and take a step, each agent receives natural language descriptions of current game state as well as observations within a predefined `{vision_radius}`. Each agent observes entities located within this local neighborhood around it, while regions outside the vision radius remain unobserved. The observed information contains the following:

- **Turn Counter:** current turn index and episode horizon (Turn t out of total turn T).
- **Agent State:** the agent’s grid position, cumulative reward, health points remaining, and attack cooldown timer (number of turns before an agent can attack again).
- **Inventory:** counts of participated kills.
- **Orientation:** current facing direction.
- **Beam Tiles Position:** the grid ceels in n the agent’s facing direction that are targeted by the beam attack when the agent uses the beam attack.
- **Nearby Agents (within `vision_radius`):** for each visible agent, its coordinates, relative offset (e.g., up 1, right 2), and a trust score from the reputation module; additionally indicates whether any agent is adjacent (Manhattan distance 1).

- **Nearby Obstacles (within `vision_radius`):** coordinates of walls or unpassable tiles with relative offsets.
- **HITTABLE_TARGETS:** positions of all resources, if any, that are within beam length in forward direction.
- **ATTACK_VALID:** whether there are hittable targets within vision distance.
- **Nearby Resources (within `vision_radius`):** coordinates of visible resources for attack with relative offsets.
- **Attackable Resources (within `beam_length`):** resource positions that lie within the forward beam up to length.
- **Action Reference:** a reminder of the action mapping.
- **Optional Tip:** when a resource is adjacent and an ally is adjacent, the observation includes a coordination hint about messaging.

A text example of the observation an agent receives:

```

CURRENT STATE:
Turn 0/50
Your position: (1, 1)
Your cumulative reward: 0.0
Your health: 12/12
Attack cooldown: 0
Inventory: hare=0, stag=0
Facing direction (beam): north
ORIENTATION: NORTH
BEAM_LEN: 3
BEAM_TILES_RC: [(0, 1), (-1, 1), (-2, 1)]
Nearby agents within vision: none
Nearby walls/unpassable tiles: (0, 0) [up 1, left 1], ...
HITTABLE_TARGETS: []
ATTACK_VALID: false

```

```
Hares within vision: (1, 6) [right 5], (5, 5) [down 4, right 4]
Hares in beam: none
Stags within vision: (3, 5) [down 2, right 4]
Stags in beam: none
Actions: 0=stay, 1=up, 2=right, 3=down, 4=left, 5=attack
```

3.1.5 Memory

Each agent maintains an explicit internal memory system that persists information across turns (when each agent takes a step in the same) and episodes (when the game ends). The memory for each agent is updated after every environment interaction and injected into the agent’s prompt at decision time. The memory system is composed of two complementary components: *step-level memory* and *episodic memory*.

Step-level Memory. This memory records interaction data at every turn. Each entry contains:

- **Current Environment State:** the agent’s observation at the time the action is selected as defined in [Section 3.1.4](#).
- **Selected Action:** the action id chosen by the agent at that turn based on the current environment state.
- **Observed reward:** the reward after the selected action is executed.
- **Next Environment State:** the agent’s observation after the environment transitions in response to the selected action.
- **Episode Termination:** the termination flag indicates whether the environment reached a terminal condition after the action is executed (e.g., maximum number of turns reached or task completion).
- **Metadata:** metadata includes prompt received by the agent, the LLM model response, and parsed reasoning.

This buffer serves as a raw interaction log that captures the structure of agent-environment dy-

namics. This memory is not directly injected into the model prompt at every turn. It supports analysis, debugging, and downstream learning and fine-tuning agents if enabled.

Episodic Memory. In parallel to the step-level memory, each agent maintains a separate memory chain to store high-level summaries. Episodic memory contains two elements: *step notes* and *reflections*. Step notes records summary of each turn: the turn index, the chosen action, a natural language summary of the agent’s observation or reasoning, the immediate reward, and optional metadata. Reflections are short lessons extracted from past experience per episode (after each game ends) that acts as abstractions over repeated outcomes. Prompts for generating per episode reflection is in [Figure 12](#). Each turn, the agents receive the last k step summaries plus the latest reflection as past experience and reflection to help guide their decision making.

3.1.6 Messages

Agents have the options to send messages to each other when they are within certain distance and both encounter potential resources to attack. Each turn, each agent check their inbox for any messages received by other agents, (agent id, message string) are stored and appended to agent’s observation text as context to guide the agent’s decisions for next turn.

3.1.7 Reputation

We introduce reputation as a trust signal that captures an agent’s belief about the reliability of its partners. In repeated coordination settings, such beliefs are necessary for cooperation, as agents decide who to trust when coordination is risky.

Specifically, reputation is a trust score per pair of agents. For each agent, reputation stores a dictionary with key (me, other) and value being a score. The score is thus directional: trust from agent i to agent j is stored and updated independently from trust in the opposite direction. All reputation values are initialized neutrally and updated incrementally over time.

Reputation updates are grounded primarily in observed behavior and coordination outcomes, rather than language alone. After each interaction, an agent updates its trust toward another agent

based on whether the other agent’s actions align with the communication intent and whether coordination succeeded in practice. In particular, reputation updates include the following:

- **Commitment Follow-through:** if an agent commits (through message) to a specific action (e.g. cooperate on stag) and a subsequently executes that action, trust score increases. If the agent fails to act according to their message will result in a trust penalty.
- **Coordination Outcomes:** successful joint outcomes such as receiving rewards from coordinate on stag will produce positive trust updates.
- **Free-riding:** agents that are nearby and able to participate in coordination but do not attack receive trust penalties.

The reputation scores are applied as incrementing deltas. Large magnitude updates are for positive outcomes, while purely communicative signals have limited or no effect unless supported by action. At each step, the reputation scores are incorporated to the model as part of the observation context. These trust signals influence agent choices such as which nearby agents to rely on for cooperation and whether to attempt coordinations with high risk.

3.2 Environment Design

This section presents game environment. We mainly discuss staghunt game setting.

In the traditional Stag Hunt game, two players simultaneously choose whether to cooperate (hunt stag) or defect (hunt hare). The payoffs are defined by a static matrix: mutual cooperation yields the highest collective reward, defection is individually safe but socially suboptimal, and failed coordination leads to low payoff.

In our setting, the payoff structure is grounded in a physical, location-based environment. We define agents in a shared gridworld and they must navigate, orient, and act in space to obtain rewards.

3.2.1 Physical Staghunt Game Description

In our gridworld implementation of the Stag Hunt game, two or more agents navigate a shared map containing hares, stags, and obstacles. They choose between safe solo rewards and higher-value cooperative targets. Each turn, agents select an action (moving, staying, or attacking by firing a forward-facing beam). Hares have low health and can be defeated alone for a small reward, while stags have higher health and can be attacked cooperatively to yield a large shared reward. Attacking costs health and may incur penalties if no target is hit to encourage efficient coordination. Agents may communicate with other visible agents to coordinate stag hunts. In addition, after a resource is killed, the respawn mechanism may regenerate the same resource type with a certain probability. Pursuing hares offers guaranteed but low payoffs, while going for stags requires trust and timing. The environment parameters such as reward, health points, attack range, and cooldowns can be modified to test robustness across configurations.

3.2.2 Game Configuration

Layers. We include three layers as follows.

- **Terrain:** it defines static structure of the world such as “wall” and “sand”.
- **Dynamic:** it defines movable entities like agents and resources. This layer is reset and repopulated each episode.
- **Beam:** it defines beam effects where agents shoot `AttackBeam` for attacking resources.

Entities. We list all entities in the environment below:

- **Empty:** it is a traversable cell on the dynamic layer and it can spawn resources during regeneration.
- **Wall:** it is impassable cell on terrain layer and it blocks movement and beams.
- **Spawn:** it is a passable cell on terrain layer for agent spawn with no rewards
- **Sand:** it is a terrain cell that allows resource respawn.
- **StagResource:** it is the stag with high reward value. Its health point decreases if attacked by

agents.

- **HareResource:** it is the hare with lower reward value. Its health point decreases if attacked by agents.
- **AttackBeam:** it is a beam on the beam layer for attacking resources. It disappears after 1 turn.

Gridworld Generation. To generate gridworld environment, we first define grid dimensions $H \times W$. Each entity is mapped to a unique symbolic representation. We then initialize the positions of all entities in the gridworld based on predefined layers and rules. This yields the initial grid world configuration. After each turn, the environment is transitioned for the next iteration.

With this physical setting, cooperation is no longer an explicit action but an emergent outcome of spatial proximity, timing, and aligned behavior. As a result, it allows us to study cooperation as a dynamical process rather than a strategic choice.

4 Experiment

4.1 Goal

The primary goal of our experiments is to assess whether LLM agents coordinate by reasoning over environment dynamics and game rules, or whether their behavior is driven primarily by surface-level symbolic cues that activate prior knowledge about the game.

4.2 Models

We evaluate five LLM agents as zero-shot agents in multi-agent environments: GPT-4o, Qwen2.5-7B-Instruct, Mistral-7B-Instruct-v0.3, LLaMA-3.1-8B-Instruct, and Gemma-2-9B-Instruct. All agents are instantiated from these models without any fine-tuning or gradient updates, and act solely based on their pretrained weights and input prompts. In each experiment, all agents within a group are instances of the same model.

4.3 Setup

Each experiment consists of a multi-agent game involving agents instantiated from the same model. Agents interact in environments defined by textual descriptions of the rules, actions, and payoff structure. All games are turn-based and require agents to make discrete decisions (e.g., select an action, coordinate with another agent). All agents receive the same rule specification. The agents do not receive any task-specific fine-tuning or gradient-based learning. At each decision point, the model returns a structured response indicating its chosen action and a natural language justification.

To test generalization beyond symbolic cues, each game is evaluated under two conditions:

- **Natural Framing:** Game elements use familiar, semantically meaningful terms (e.g., “stag,” “hare”).
- **Neutral Framing:** All symbols and entities are renamed using abstract or invented terms (e.g., “ijjhu,” “guydguug”), while preserving the underlying reward and coordination structure.

This setup allows us to do controlled comparisons of model behavior across structurally identical but semantically different environments.

Each parameter combination is run for predefined number of episodes, depending on computational resources. All actions, messages, and reasoning traces are logged for later analysis. Experiment metrics for evaluation are computed per model and per game, as described in [Section 4.4](#).

4.4 Metrics

[Table 2](#) shows five evaluation metrics. We compare each of them under both natural and neutral framings.

First Successful Cooperation. This metric measures the number of turns until a cooperative outcome is first achieved. For each agent pair, we record whether and when mutual coordination occurs.

Metric	What It Shows
First Successful Cooperation	How quickly agents discover and execute a successful cooperative action. The lower, the better.
Total Rewards	Measures overall efficiency. Indicates whether agents achieve high-payoff outcomes individually or as a group, relative to the game’s maximum possible reward.
Number of Stags Hunted	Tracks the number of stags hunted per game per agent.
Communication Language	Analyzes message content among agents in multi-agent games with communication. Captures trust, fairness, coordination attempts, and reliance on symbolic terms.
Reasoning Analysis	Analyzes the reasoning LLMs use to justify actions. Assesses depth and abstraction of agent understanding.

Table 2: Summary of Evaluation Metrics.

Total Rewards. Total reward measures the average cumulative payoff obtained by each agent per episode. For group games, we also report average team reward.

$$\text{TotalReward}_m = \frac{1}{N \times |A|} \sum_{i=1}^N \sum_{a \in A} R_{i,a}, \quad \text{for all } m,$$

where m is the model, A represents all agents in each group, N is the number of episodes, and $R_{i,a}$ is the reward for agent a in episode i .

Number of Stags Hunted. This metric computes the number of stags hunted per agent across multiple episodes. We plot the number of stags each agent hunts in each episode.

Communication Analysis. For environments with communication, we analyze agent messages using a rubric-based autorater (GPT4.1). The communication score measures the degree to which language provides actionable coordination signals. Scores are aggregated across time steps to produce episodic communication score. The prompt to the autorater for communication analysis is in [Figure 13](#).

Reasoning Analysis. Similar to communication analysis, we quantify We analyze agents’ chain-of-thoughts or action justification with GPT4.1. The reasoning score measures the extent to which

decisions are justified through grounded, structural reasoning rather than symbolic or normative cues. The prompt to the autorater for reasoning analysis is in [Figure 14](#).

4.5 Results

We evaluate different models across multiple tasks with evaluation metrics defined in [Table 2](#).

4.5.1 Task 1: Simple Experiment

We construct a simple experiment to test the comparative performance of different models under identical settings. [Figure 2](#) visualizes the simple game environment.

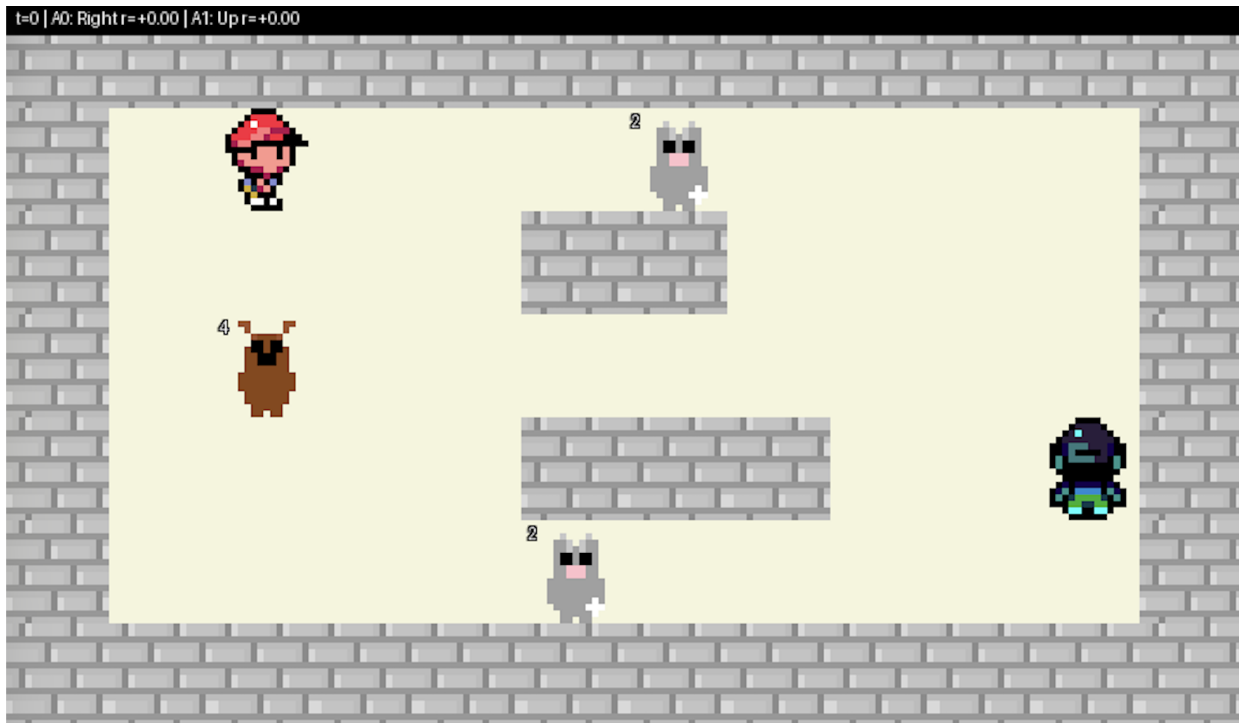


Figure 2: Simple Game Visualization. 2 agents are randomly initialized in the 7×12 Gridworld. It contains 2 hare resources and 1 stag resource. Each hare has 2 HPs and the stag has 4 HPs.

Setup. We run staghunt game in the setting for 5 episodes. Each episode runs for 50 turns. We use natural framing for both agents. We present hyperparameters of the environment in [Table 3](#).

Evaluation. We present and visualize the evaluation results in [Table 4](#), [Figure 3](#), and [Figure 4](#). Evaluation results show that GPT-4o agents obtain the highest average reward and highest rea-

Hyperparameter	Value
Vision radius	5
Beam length	3
Agent health	12
Stag reward	5
Hare reward	1
Stag health	4
Hare health	2
Stag regeneration cooldown	5
Hare regeneration cooldown	5
Attack cost	0.05

Table 3: Environment Hyperparameters in the Simple Experiment.

soning score per agent. Moreover, most models do not follow communication instructions from the prompts. In other words, both agents choose not to hunt stag cooperatively, resulting in only one agent getting full stag reward for a successful hunt. Gemma-9B, LLaMA-8B, and Mixtral-7B model show imbalance between agents as only a single agent consistently succeeds in hunting stag across multiple episodes as illustrated in Figure 3.

Model	Avg. Reward	Reasoning	Communication
Gemma2-9B	1.3	3.2	-
GPT-4o	2.9	3.8	2.5
LLaMA3.1-8B	1.7	2.8	-
Qwen2.5-7B-Instruct	2.4	3.2	-
Mixtral-7B-Instruct-v0.3	1.6	3.5	-

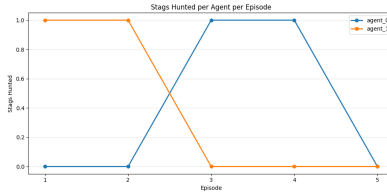
Table 4: Numerical Result of the Simple Experiment. For each model, we record average reward per agent per episode, average reasoning score per agent, and average communication score per agent. Only GPT-4o exhibits communication signals between agents.

4.5.2 Task 2: All Hare Experiment

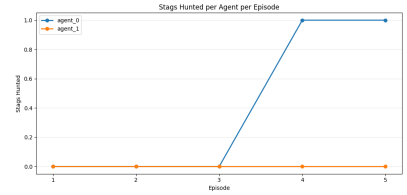
We construct the experiment where the environment contains all hares to test whether each model can effectively learn and exploit the simplest reward structure. We evaluate the performance of agents when there is no need for cooperation and communication. Figure 5 visualizes the all hare experiment.



(a) Gemma2-9B



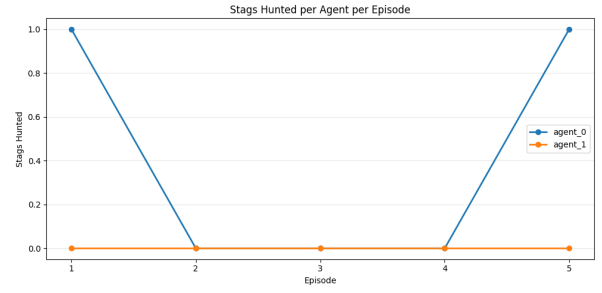
(b) GPT-4o



(c) LLaMA3.1-8B

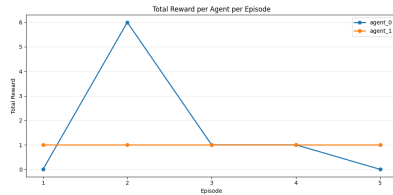


(d) Qwen2.5-7B-Instruct

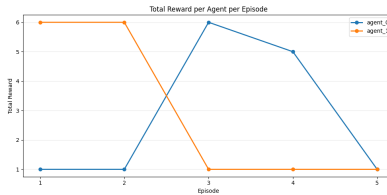


(e) Mixtral-7B-Instruct-v0.3

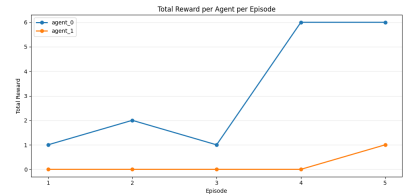
Figure 3: Number of Stags Hunted Per Episode Across Different Models.



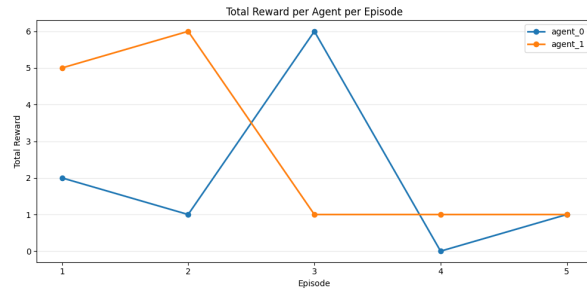
(a) Gemma2-9B



(b) GPT-4o



(c) LLaMA3.1-8B



(d) Qwen2.5-7B-Instruct



(e) Mixtral-7B-Instruct-v0.3

Figure 4: Total Reward Per Episode Across Different Models.

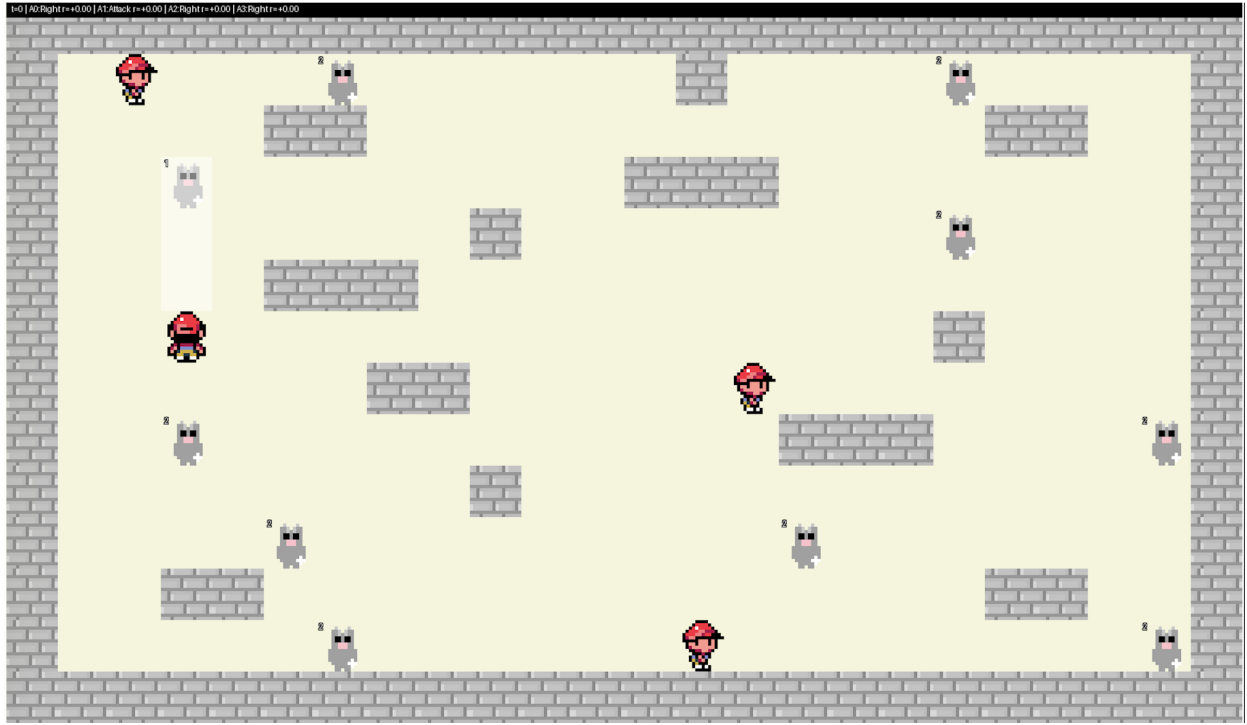


Figure 5: All Hare Game Visualization. 4 agents are randomly initialized in the 14×24 Gridworld. It contains 10 hare resources but no stag resource. Each hare has 2 HPs.

Setup. For each model, we run the all-hare experiment until all hares are eliminated or until maximum number of turns (150) is reached. Specifically, agents are allowed to attack hares and stags by themselves. If agents solo attack stags, they do not get rewards, but if they collaborate on stags, agents can split rewards. We compare performance metrics across natural framing and neutral framing, and report the number of episodes required to eliminate all hares under each framing condition. Hyperparameters of this setting is the same as the simple experiment in [Table 3](#).

Evaluation. We evaluate model performances on all-hare game across natural framing and neutral framing. Since there is no stags, there is no necessary need for communication between agents. [Table 5](#) and [Table 6](#) shows comparative results in natural framing and neutral framing, respectively. In natural framing, for one game, [Table 5](#) illustrates that GPT4o agents successfully obtain all resources in only 41 turns, receive all most full average reward, and achieve the highest average reasoning score. Specifically, one GPT4o agent receives a negative penalty by firing a beam when the resource is not within beam length. Furthermore, all 4 GPT4o agents obtain a

reasoning score of 4. Other agents such as LLaMA and Mistral models take longer time to finish all resources. They have lower average reasoning scores, with noticeable variation across agents, even though there is no miss-attack penalties. Agents like Gemma and Qwen do not obtain full reward within 150 steps, but they have relatively high reasoning scores. The comparison across models under the same framing illustrate that model size is not the determinant factor in model performance (the average reward obtained by all agents). Training data (incorporating more conversational data), model structure, and training methods (trained with human feedback) are all factors that potentially influence results and observations. For example, Qwen model is specifically tuned for instruction-following and (math) logical reasoning, suggesting higher reasoning score, but it might not translate them into concrete actions in the multi-agent setup. On the other hand, Mixtral is a Mixture-of-Expert model where as other models are dense transformers. This means there are experts specializing different tasks which can lead to strong performance.

When comparing between the two framings, we see that most models perform better under neutral framing. In other words, under natural framing, agents either finish early, or obtain higher average rewards. This shows that that prior knowledge in symbolic cues do provide certain amount of additional information that affects agents’ behaviors, and that LLM agents do not generalize well to structurally identical tasks when symbolic meaning is removed.

Model	Turns till Finish	Avg. Reward	Avg. Reasoning
Gemma2-9B	150	2.0	3.5
GPT-4o	41	2.4895	4.0
LLaMA3.1-8B	108	2.5	3.0
Qwen2.5-7B-Instruct	150	2.25	3.25
Mixtral-7B-Instruct-v0.3	138	2.5	3.75

Table 5: Numerical Result of the Allhare Experiment with **Natural Framing**. For each model, we record number of turns till the all hares have been killed by all agents or when maximum number of turns is reached (150), average reward per agent per episode, and average reasoning score per agent.

4.5.3 Task 3: Complicated Experiment

We construct a more complicated experiment with more obstacles and more resources to test the ability for agents to navigate a more complex environment. Moreover, we evaluate whether the

Model	Turns till Finish	Avg. Reward	Avg. Reasoning
Gemma2-9B	150	1.7375	3.75
GPT-4o	63	2.5	4.0
LLaMA3.1-8B	150	2.25	2.75
Qwen2.5-7B-Instruct	150	2.25	4.0
Mixtral-7B-Instruct-v0.3	150	2.25	3.5

Table 6: Numerical Result of the Allhare Experiment with **Neutral Framing**. For each model, we record number of turns till the all hares have been killed by all agents or when maximum number of turns is reached (150), average reward per agent per episode, and average reasoning score per agent.

agents’ own reflections after a game help improve their future performances. [Figure 12](#) shows the prompt to GPT4.1 for generating reflection after a game.

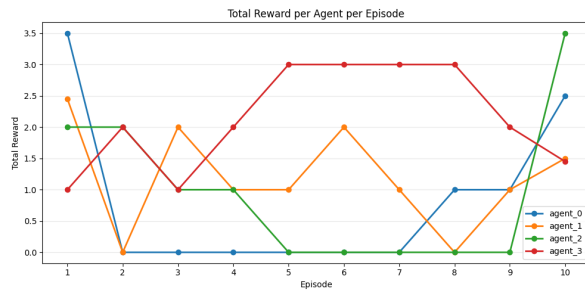
Setup. We fix the prompt framing to natural framing. We evaluate on GPT4o model only due to computational limitations. We compare non-reflective agents, which do not generate post-game self-reflections, and reflective agents, which generate a reflection after each episode before continuing to the next one. The primary comparison uses 10 episodes for both variants to isolate the effect of reflection. To further study whether reflection yields cumulative benefits over longer interaction horizons, we additionally evaluate the reflective variant for 50 episodes. Thus, for each model, we report results for three configurations: no reflection for 10 episodes, reflection for 10 episodes, and reflection for 50 episodes. [Figure 6](#) shows a visualization of the game. Hyperparameters are in [Table 3](#).

Evaluation. We plot total rewards and stags hunted per agent for the 3 experiments.

Reflective Vs. Non-reflective Agents. [Figure 7](#) and [Figure 8](#) show the comparison between reflexive and non-reflexive agents. It illustrates reflection improves behavioral consistency but may reduce adaptability. In other words, for non-reflexive agents, there is more oscillations as shown in [Figure 8](#), implying more inconsistent strategies across games, where as reflexive agents show more consistency as shown in [Figure 7](#). Results indicate that is a trade-off between consistency and adaptability in LLM-based multi-agent systems.



Figure 6: Complicated Game Visualization. 4 agents are randomly initialized in the 14×24 Gridworld. It contains 6 hare resources and 4 stag resources. Each hare has 2 HPs, and each hare has 4 HPs.

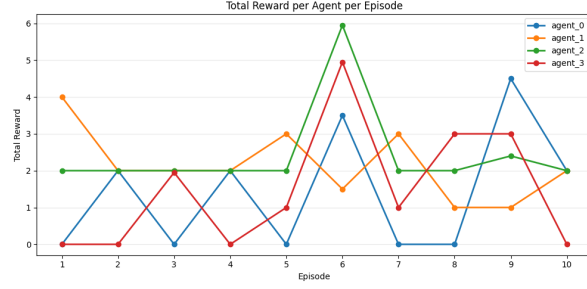


(a) Reflexive Agents Rewards



(b) Reflexive Agents Stags

Figure 7: Reflexive GPT4o Agents Playing 10 Games.

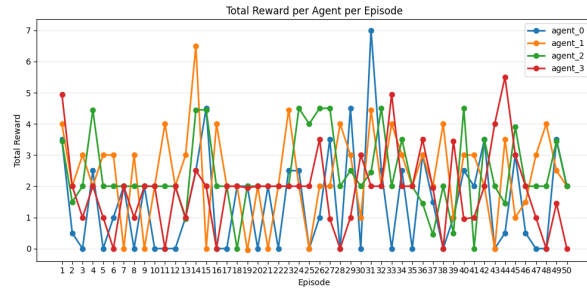


(a) Non-reflexive Agents Rewards

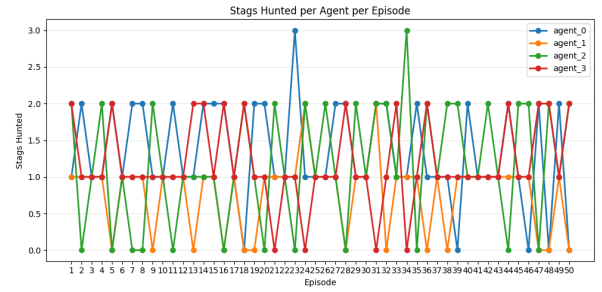


(b) Non-reflexive Agents Stags

Figure 8: Non-Reflexive GPT4o Agents Playing 10 Games.



(a) Reflexive Agents Rewards



(b) Reflexive Agents Stags

Figure 9: Reflexive GPT4o Agents Playing 50 Games.

Reflection Over Shorter Vs. Longer Period of Time. We compare reflective agents over shorter (10 games) and longer (50 games) period of time. From [Figure 7](#) and [Figure 9](#), we observe that further interaction enables agents to refine strategies, leading to higher peak rewards and more adaptive coordination patterns. This suggests that reflection serves as a form of cumulative in-context learning, rather than merely a stabilizing mechanism. Moreover, from [Table 7](#) and [Table 8](#), an overall enhancement is that rewards are more evenly distributed across agents, implying enhanced cooperation over longer term. Average reasoning and communication scores also improve significantly across all agents over longer term. Furthermore, [Table 9](#) shows improvements in average reward per agent and first successful cooperation per game (the earliest timestep in an episode where at least two agents coordinate to hunt a stag, i.e, the lower, the better) when agents generate reflections for longer time.

Agent	Avg. Reward	Avg. Reasoning	Avg. Communication
1	0.80	3.1	2.0
2	1.20	3.0	-
3	0.95	3.2	3.0
4	2.05	3.3	-

Table 7: Reflexive Agents Result Playing 10 Games.

Agent	Avg. Reward	Avg. Reasoning	Avg. Communication
1	1.30	3.94	2.89
2	1.05	3.82	2.69
3	1.55	3.86	2.81
4	1.85	3.62	2.84

Table 8: Reflexive Agents Result Playing 50 Games.

Number of Games	Avg. Reward Per Agent	Avg. First Successful Cooperation Per Game
10	1.27	71
50	2.04	17

Table 9: Comparison of Reflexive Agents for Shorter and Longer Time.

[Section 6.2](#) shows the reflections for all 4 agents across 10 games. It is important to note that although the feedback emphasize coordination and strategic exploration, they are really repetitive and general in that the verbal reflection alone does not provide sufficient signal for effective policy

improvement in multi-agent environments. This matches with our observations that we do not see significant reward improvement comparing reflection-based agents and non-reflection-based agents. However, reflection does help when agents are playing games for longer period of time.

5 Conclusion and Future Work

In this work, we present a controlled multi-agent benchmark to evaluate whether LLM agents perform true incentive-based reasoning or depend on symbolic cues. By using a gridworld formulation of the Stag Hunt game, we vary environment framing, complexity, and agent capabilities to analyze coordination behavior across models.

Our findings reveal three key insights. First, LLM agents exhibit strong sensitivity to linguistic framing: performance degrades when semantic cues are removed, indicating that agents often rely on prior associations rather than abstract reasoning over payoff structures. Second, in some environments, most models struggle to achieve stable cooperation, with coordination often emerging inconsistently or asymmetrically across agents. Third, reflection mechanisms introduce a trade-off between stability and adaptability: while reflection improves long-term coordination and behavioral consistency, it can reduce exploration and short-term reward efficiency.

Overall, these results suggest that current LLM agents do not fully internalize underlying incentive structures in multi-agent settings. Instead, their behavior is heavily influenced by prompt semantics and prior knowledge. This highlights a key limitation in zero-shot generalization and raises important questions about the robustness of LLM-based agents in real-world decision-making environments.

Future work can investigate more structured and robust reflection generation, extend evaluation to more diverse OOD scenarios, encourage emergent communication protocols, and explore generalization on other social environments. Furthermore, a natural next step is to analyze raw reasoning traces to identify the specific strategies used by LLM agents, providing deeper insight into whether they acquire reusable coordination policies in-context or simply rely on scenario-specific pattern matching based on symbolic cues.

Limitations. Beyond computational constraints that limit model scale, our study has several additional limitations. First, all agents operate in a zero-shot setting without learning across episodes, which may not reflect adaptive behaviors in more realistic settings. Second, experiments are restricted to a gridworld Stag Hunt environment, limiting generalization to other social dilemmas or real-world tasks. Third, communication is simplified and reasoning quality is evaluated using an LLM-based autorater, which may introduce bias. Finally, the experimental setup focuses on limited interaction horizons and controlled environments with explicit rules.

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6 Appendix

6.1 Prompts

Agent Prompt

Role: strategic agent

You are an autonomous agent acting in a multi-agent decision process. Multiple agents act simultaneously each turn. Your goal is to maximize reward according to the environment's rules.

Input You Receive Each Turn:

- Observation (state/features visible to you)
- Recent Messages and History
- Reputation (compact stats you can rely on)
- Actions Available (what you can do)

Game Mechanics

`vision_radius`: you only observe tiles within this radius

`beam_length`: attack reach

Orientation and Attack Geometry

Attack fires a beam forward in your current facing direction.

- The beam can only hit a target *directly in front of you* and *within beam length*.
- A tile is considered *in front of you* if and only if:
 1. facing north: same column, smaller row index
 2. facing south: same column, larger row index
 3. facing east: same row, larger column index
 4. facing west: same row, smaller column index
- If no valid target is in front of you or in the beam, the attack is *wasted* and may be *penalized*.
- Targets that are behind you, to your side, diagonal, or outside beam length are *not*

hittable.

Use the computed fields in the Observation (BEAM_TILES_RC, HITTABLE_TARGETS, ATTACK_VALID). Do not recompute geometry.

Health and Rewards

- You and targets have *health points (HP)*.
- An attack can reduce *target HP* and also reduce *your HP*.
- Rewards are tied to the environment's reward rules. You can only receive rewards when a target's HP reaches 0.
- Attack damage to resources persists, but resources may regenerate HP if enough time has passed since last attack.
- You cannot attack if your HP reaches 0.

Default Action Policy (General, Not Direction-Specific)

- If ATTACK_VALID is true, attack is permitted. Attack only if beneficial under the reward rules and coordination context (e.g., target can be completed alone, or allies are nearby/aligned).
- If targets are visible but attacking now is not beneficial: move to reduce distance, improve beam alignment, or increase coordination likelihood.
- If no targets are visible: explore to gather information, avoiding repeated ineffective moves.

Mandatory Per-Turn Reasoning Procedure

Before choosing an action, you must internally perform:

1. Identify your current *position* and *orientation*.
2. Read ATTACK_VALID, HITTABLE_TARGETS, and coordination signals from the Observation.
3. If ATTACK_VALID is true: decide whether attacking now is advantageous given rewards, HP, and coordination.
4. If ATTACK_VALID is false and targets are visible: choose movement improving future attack feasibility or coordination, avoiding unpassable tiles.

5. If no targets are visible: explore systematically and avoid repeating no-progress actions.
6. Choose exactly one action from the available action set.

Communication

If "Nearby agents within vision" is NOT "none", you should usually send a short message (≤ 5 words) indicating your intent in the MESSAGE field. Use messaging only if it might affect or be affected by another agent's decision; otherwise leave it empty.

Reward Summary: {reward_rule}

Current Turn

Observation: {obs_text}

Memory & Context: {memory_text}

Action Available: {action_table}

- a = 0: stay, your position and orientation does not change
- a = 1: up, you move up (y axis decreases by 1) and orientation does not change
- a = 2: right, you move right (x axis increases by 1) and orientation becomes east
- a = 3: down, you move down (y axis increases by 1) and orientation becomes south
- a = 4: left, you move left (x axis decreases by 1) and orientation becomes west
- a = 5: attack, you do not move, you fire a beam in your current orientation, beam only hits targets directly in front of you and within beam length

Response Format

Return ONLY valid JSON following this structure:

```
{
  "REASONING": string,    // 1 to 2 concise sentences,
  "ACTION": number,       // integer action id from {action_table},
  "MESSAGE": string,      // keep within 5 words
  "CONFIDENCE": number    // integer between 0 and 100
}
```

Constraints

- REASONING: 1–2 concise sentences. Must mention: current orientation, whether attack is valid, the chosen target (if any) and its remaining HP, whether there is any walls/unpassables that block you, why the chosen action is appropriate now
- ACTION: must be a valid action ID from the action table
- MESSAGE: empty string if no ally
- CONFIDENCE: integer from 0 to 100

Attention

1. Only return a valid JSON object with the four required fields.
2. Do not include any explanations or natural language text.
3. Ensure the format strictly matches the example above.
4. Do not invent entities or actions.
5. Use only information provided in Observation, Recent Messages and History, and Reputation.

Your Response:

Figure 10: General Agent Prompt.

6.2 Reflections

Below is the reflection generated by the 4 GPT4o agents (agents 0 – 3 respectively) in 10 consecutive games.

Listing 1: Episode 1 Reflections.

```
{
  "0": "In this episode, I focused heavily on exploration, resulting in limited attacks and no stag captures. In future episodes, I should prioritize better coordination with nearby agents for stag hunting and optimize my exploration to encounter more targets.",
  "1": "In this episode, my exploration strategy was hindered by walls, leading to repeated ineffective moves and missed opportunities to
```

```

    find targets. In future episodes, I should prioritize mapping out
    the environment more efficiently and focus on coordinating with
    other agents to target stags for higher rewards.",
"2": "In this episode, despite frequent exploration, I struggled to
    find and effectively engage targets, resulting in minimal rewards.
    In future episodes, I'll focus on optimizing my movement strategy
    to prioritize areas with higher target likelihood and improve
    coordination with other agents to increase the chances of
    successful stag hunts.",
"3": "In this episode, I learned that my exploration strategy often
    led to repetitive and ineffective movements, which limited my
    ability to find and engage targets. In future episodes, I'll focus
    on more systematic exploration and better utilize communication
    to coordinate with other agents for stag hunts, maximizing
    potential rewards."
}

```

Listing 2: Episode 2 Reflections.

```

{
    "0": "In this episode, I realized that my exploration was too
        repetitive and lacked effective coordination with other agents,
        leading to limited encounters with targets. In future episodes, I
        should focus on strategic exploration to locate more resources and
        improve communication for coordinated stag hunting.",
    "1": "In this episode, I learned that inefficient exploration due to
        walls led to missed opportunities. In future episodes, I should
        focus on better mapping the environment and coordinating with
        other agents to target stags for higher rewards.",
    "2": "In this episode, I learned that my exploration strategy was
        inefficient, leading to minimal target engagement and rewards. In
        future episodes, I will prioritize exploring areas with higher

```

```

    target likelihood and enhance coordination with other agents to
    improve the success rate of stag hunts.",
    "3": "In this episode, I learned that my exploration strategy was
    often repetitive and ineffective, limiting target engagement. In
    future episodes, I'll focus on more systematic exploration and
    utilize communication to coordinate with other agents, especially
    for stag hunts, to maximize rewards."
}

```

Listing 3: Episode 3 Reflections.

```

{
    "0": "In this episode, I learned that my exploration was too
    repetitive and lacked effective coordination, which limited target
    encounters. In future episodes, I should focus on more strategic
    exploration and improve communication with other agents for better
    coordination, especially in stag hunting.",
    "1": "In this episode, I learned that inefficient exploration and
    lack of coordination limited my ability to target stags for higher
    rewards. In future episodes, I should focus on better mapping the
    environment and collaborating with other agents for more
    effective stag hunts.",
    "2": "In this episode, I realized my exploration strategy was
    inefficient, limiting target engagements and rewards. To improve,
    I will focus on exploring high-target areas and enhancing
    coordination with other agents, especially for stag hunts.",
    "3": "In this episode, I realized that my exploration strategy was
    too repetitive and limited my engagement with targets. In future
    episodes, I'll aim for more systematic exploration and use
    communication to better coordinate with other agents, especially
    for stag hunts, to maximize rewards."
}

```

Listing 4: Episode 4 Reflections.

```
{
  "0": "In this episode, I learned that my exploration was too
        repetitive and lacked effective coordination, which limited target
        encounters. In future episodes, I should focus on more strategic
        exploration and improve communication with other agents for better
        coordination, especially in stag hunting.",
  "1": "In this episode, I realized that inefficient exploration and
        insufficient coordination hindered my ability to target stags for
        higher rewards. Moving forward, I should prioritize better
        environment mapping and collaborate more effectively with other
        agents to maximize reward opportunities, particularly for stag
        hunts.",
  "2": "In this episode, I learned that my exploration strategy was
        inefficient, limiting my ability to engage with targets and
        maximize rewards. In future episodes, I should focus on exploring
        high-target areas more effectively and enhancing coordination with
        other agents, particularly for stag hunts.",
  "3": "In this episode, I learned that my exploration strategy was too
        repetitive, which limited my engagement with targets. In future
        episodes, I'll focus on more systematic exploration and improve
        communication with other agents, especially for coordinating stag
        hunts to maximize rewards."
}
```

Listing 5: Episode 5 Reflections.

```
{
  "0": "In this episode, I realized that my exploration was overly
        repetitive and lacked coordination, which limited encounters with
        targets. In future episodes, I should focus on more strategic
        exploration, diversify my movement patterns, and enhance
```

```

communication with other agents to improve coordination,
especially for stag hunting.",
"1": "In this episode, I learned that optimizing exploration and
enhancing coordination with other agents are crucial for
maximizing rewards, especially for stag hunts. In future episodes,
I will focus on mapping the environment more efficiently and
collaborating effectively to target high-reward opportunities.",
"2": "In this episode, I learned that my exploration strategy was
inefficient, which limited my engagement with targets and rewards.
In future episodes, I should prioritize exploring areas with
higher target likelihood and improve coordination with other
agents, especially for stag hunts.",
"3": "In this episode, I realized that my exploration strategy was
too repetitive, limiting my engagement with targets. In future
episodes, I'll focus on more systematic exploration and enhance
communication with other agents, especially for coordinating stag
hunts to maximize rewards."
}

```

Listing 6: Episode 6 Reflections.

```

{
"0": "In this episode, I learned that my exploration was too
repetitive and lacked effective coordination, which limited my
encounters with targets. In future episodes, I should diversify my
movement patterns and enhance communication with other agents to
improve coordination, especially for stag hunting.",
"1": "In this episode, I learned the importance of efficient
exploration and coordination for maximizing rewards, particularly
in stag hunts. In future episodes, I'll focus on better mapping
the environment and collaborating with other agents to target high
-reward opportunities effectively.",

```

```

"2": "In this episode, I learned that my exploration strategy was
      inefficient, limiting target engagement and rewards. In future
      episodes, I should focus on exploring areas with higher target
      likelihood and enhance coordination with other agents,
      particularly for stag hunts.",
"3": "In this episode, I learned that my exploration strategy was too
      repetitive, which limited my engagement with targets. In future
      episodes, I will focus on more systematic exploration and enhance
      communication with other agents, especially for coordinating stag
      hunts to maximize rewards."
}

```

Listing 7: Episode 7 Reflections.

```

{
  "0": "In this episode, I learned that my exploration was too
        repetitive and lacked effective coordination, limiting my
        encounters with targets. In future episodes, I should diversify my
        movement patterns and enhance communication with other agents to
        improve coordination, especially for stag hunting.",
  "1": "In this episode, I learned the critical value of strategic
        exploration and effective coordination with other agents,
        especially for high-reward stag hunts. In future episodes, I'll
        prioritize mapping the environment more efficiently and
        collaborating closely with allies to maximize shared rewards.",
  "2": "In this episode, I realized that my exploration strategy was
        inefficient, which limited my engagement with targets and rewards.
        In future episodes, I should prioritize exploring high-target
        areas and improve coordination with other agents, especially for
        stag hunts, to maximize rewards.",
  "3": "In this episode, I realized that my exploration strategy was
        too repetitive, limiting target engagement. In future episodes, I

```

```
will adopt a more systematic exploration approach and enhance
communication with other agents to better coordinate stag hunts
and maximize rewards."
}
```

Listing 8: Episode 8 Reflections.

```
{
  "0": "In this episode, I learned that my movement patterns were too
    repetitive, limiting my encounters with targets. In future
    episodes, I should diversify my exploration strategy and enhance
    communication with other agents to improve coordination,
    especially for stag hunting.",
  "1": "In this episode, I learned the importance of efficient
    exploration and the need for better coordination with allies,
    especially for high-reward stag hunts. In future episodes, I'll
    focus on optimizing my movement to map the environment more
    effectively and collaborate closely with others to maximize shared
    rewards.",
  "2": "In this episode, I learned that my exploration strategy was
    inefficient, often blocked by walls, which limited my ability to
    engage with targets. In future episodes, I should prioritize
    exploring high-target areas and improve coordination with other
    agents, especially for stag hunts, to maximize rewards.",
  "3": "In this episode, I learned that my exploration strategy was too
    repetitive, which limited my engagement with targets. In future
    episodes, I will adopt a more systematic exploration strategy and
    improve communication with other agents to better coordinate stag
    hunts and maximize rewards."
}
```

Listing 9: Episode 9 Reflections.

```

{
  "0": "In this episode, I realized that my exploration strategy was
        too repetitive, reducing opportunities to encounter targets. In
        future episodes, I should diversify my movement patterns and
        enhance communication with other agents to improve coordination,
        especially for stag hunting.",
  "1": "In this episode, I learned that optimizing movement and
        enhancing coordination with allies are crucial for maximizing
        rewards, especially in stag hunts. In future episodes, I'll focus
        on strategic exploration and better communication to improve
        collaboration and resource acquisition.",
  "2": "In this episode, I realized that my exploration was often
        hindered by walls, limiting target engagement. To improve, I
        should focus on exploring areas with higher target potential and
        enhance coordination with other agents, particularly for stag
        hunts, to maximize rewards.",
  "3": "In this episode, I realized that my exploration was too
        repetitive, reducing my opportunities to engage with targets
        effectively. In future episodes, I'll implement a more systematic
        exploration strategy and enhance communication with other agents
        to improve coordination and maximize rewards, especially for stag
        hunts."
}

```

Listing 10: Episode 10 Reflections.

```

{
  "0": "In this episode, I learned that my exploration strategy was too
        repetitive, limiting target encounters. In future episodes, I
        should diversify movement patterns and enhance communication with
        other agents to improve coordination, particularly for stag
        hunting.",
}

```

```
"1": "In this episode, I learned the importance of strategic movement
    and effective communication for coordinating with allies,
    especially for stag hunts. In future episodes, I'll focus on
    improving exploration strategies and enhancing collaboration to
    maximize rewards and resource acquisition.",
"2": "In this episode, I learned that my exploration was often
    limited by walls, reducing my engagement with targets. In future
    episodes, I should prioritize exploring areas with higher target
    potential and improve coordination with other agents, especially
    for stag hunts, to maximize rewards.",
"3": "In this episode, I learned that my exploration strategy was too
    repetitive, limiting my engagement with targets. In future
    episodes, I'll adopt a more systematic exploration approach and
    enhance communication to improve coordination, especially for stag
    hunts."
}
```

Agent Prompt

Parameters:

```
hare_reward = {hare_reward}
stag_reward = {stag_reward}
agent_health = {agent_health}
stag_health = {stag_health}
hare_health = {hare_health}
attack_cost = {attack_cost}
regeneration_rate = {regeneration_rate}
stag_regeneration_cooldown = {stag_regeneration_cooldown}
hare_regeneration_cooldown = {hare_regeneration_cooldown}
```

Reward Rules:

- When you choose to attack, the beam fires only forward and only hits targets within beam length.
- Attacking when no hare or stag is within the beam is penalized.
- Resource reward is either `hare_reward` or `stag_reward`; if multiple agents attack, the reward is split evenly.
- Each resource (hare or stag) has health points (HP). Each agent also has HP. Hare has `hare_health` HP; stag has `stag_health` HP. Each agent starts with `agent_health` HP.
- On a valid hit, hare or stag HP decreases by 1. If a resource's HP reaches 0 or below, the resource is defeated and removed; attacking agents share the reward.
- If a hare is defeated, each attacker gains `hare_reward/num_attackers` reward. A stag only pays out if at least `stag_quorum_k` distinct agents contributed to defeating it. If a stag is defeated with enough contributors, each attacker gains `stag_reward/num_attackers` reward. If fewer than `stag_quorum_k` agents contribute to a stag kill, each contributor gets `sucker_payoff`.
- Whenever you attack (even if the attack hits), your HP decreases by 1, and your reward decreases by `attack_cost`.
- If a hare or stag has HP below its maximum, it may regenerate over time. Regeneration begins after `hare_regeneration_cooldown` turns for hare and `stag_regeneration_cooldown` turns for stag since the last attack, at a rate of `regeneration_rate` HP per turn.

System Hints:

- Stag is higher value than hare, and coordination increases expected reward.
- Regeneration means partial progress can be lost, so consider focusing on finishing off targets.
- Hare is safe individually; stag requires cooperation to avoid wasted attacks.

Figure 11: Natural Framing of Stag Hunt Reward Rule.

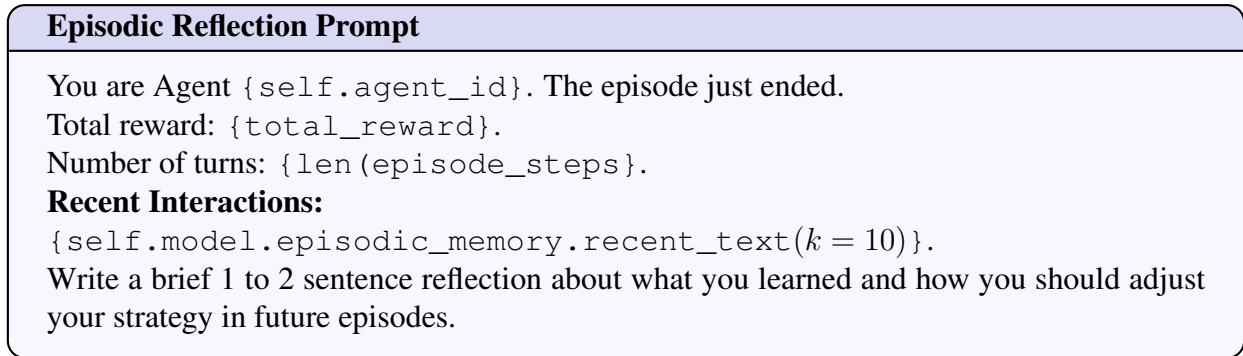


Figure 12: Reflection Prompt.

Communication Analysis Prompt

You are an impartial evaluation model. Rate the following list of short communication messages sent by an agent during a single game episode to express intent to another agent.

Attention:

- Messages are intentionally short.
- Do NOT penalize brevity.
- Evaluate how much coordination-relevant information is encoded clearly and unambiguously in the message.

Definition: The communication score measures whether the message conveys actionable coordination intent despite extreme brevity.

Agent message: The agent's list of sent message texts for the episode: {message}

Episodic-level context: {context}

Scoring scale (0–3):

- 0 = no coordination signal or purely symbolic token
- 1 = weak or ambiguous coordination cue
- 2 = clear coordination intent (who/what) but incomplete
- 3 = explicit, actionable coordination signal (clear action and target)

Evaluation criteria: When scoring, consider whether the message:

- Specifies an action, target, or role relevant to coordination
 - Disambiguates intent enough to influence another agent's decision
 - Is consistent with shared observations and environment mechanics
 - Avoids references to entities or plans not supported by context
- Score the episode based on the strongest coordination signal present in the message list.

Output format: Return ONLY valid JSON:

```
{
  "communication_score": 0-3,
  "evidence_spans": "exact words supporting the score"
}
```

Communication_score must be an integer from 0 to 3, and evidence_spans should quote the exact words from the message list that justify the score.

Figure 13: Communication Analysis Prompt.

Reasoning Analysis Prompt

You are an impartial evaluation model. Rate the following list of agent reasoning traces and action justifications produced across multiple turns in a single game episode.

Definition: The reasoning score measures the extent to which the agent justifies its action using grounded, structural reasoning about incentives, coordination requirements, and environment mechanics relative to what the agent could reasonably observe and know at that turn. The score should not reward symbolic associations, social or normative language, or generic statements about cooperation that are not supported by the environment state.

Reasoning text: A JSON list of reasoning strings produced by the agent (a list reasoning outputs for each turn): `{reasoning}`

Episodic-level context: `{context}`

Scoring scale (0–5):

- 0 = purely symbolic or normative reasoning, no grounding
- 1 = mostly symbolic with minimal reference to environment
- 2 = mixed reasoning with weak or implicit grounding
- 3 = moderate structural reasoning with some grounding
- 4 = strong structural reasoning grounded in observations or rules
- 5 = explicit, detailed reasoning about incentives, coordination, and environment mechanics

Evaluation criteria: When scoring, check whether the reasoning:

- References observable facts (positions, targets, visibility, mechanics)
- Correctly reflects environment rules (e.g., attack constraints, coordination requirements)
- Avoids claims about coordination, threats, or incentives not supported by the agent's observation
- Explains why the chosen action follows from the environment state

Score the episode based on the strongest grounded reasoning exhibited in the list.

Output format: Return ONLY valid JSON:

```
{
  "reasoning_score": 0-5,
  "evidence_spans": ["short quote 1", "short quote 2"]
}
```

Reasoning_score must be an integer from 0 to 5, and evidence_spans should quote short excerpts from reasoning that justify the score.

Figure 14: Reasoning Analysis Prompt.